Reg. No.															

B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Fifth Semester

21AIC301J - DEEP LEARNING TECHNIQUES

(For the candidates admitted from the academic year2021-2022 to 2023-2024)

Note:
(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours — Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer **ALL** Questions

	Question	Marks	BL	CO	PO
1.	Find the correct statement for the matrix A and B A) $(AB)^{-1} = B^{-1}A^{-1}$ B) $(A^T) = A$ C) AB = BA D) A*I = I	1	1	1	2
2.	Overfitting occurs because of ----------- A) high bias and low variance B) low bias and high variance C) high bias and high variance D) low bias and low variance	1	1	1	2
3.	Deep learning algorithms use ----------- layers A) 2 B) 5 C) 3 D) many numbers of	1	1	1	2
4.	Convergence in perceptron learning takes place if and only if A) a minimal error condition is satisfied B) actual output is close to desired output C) classes are linearly separable D) error value is negative	1	1	1	2
5.	Find the activation function from the given list which can lead to vanishing gradients A) ReLU B) Tanh C) Leaky ReLU D) Parameterized Relu	1	1	2	2
6.	After training a neural network, you observe a large gap between the training accuracy (100%) and the test accuracy (42%). Which of the following methods is commonly used to reduce this gap? A) Generative Adversarial Networks B) Dropout C) Sigmoid activation D) RMSprop optimization	1	2	2	2
7.	Eary stopping is a kind of ------------ A) cross-validation strategy B) co-linear technique C) performance monitoring technique D) epoch finding strategy	1	1	2	2
8.	Neural Networks are complex -------------- with many parameters A) Linear Functions B) Nonlinear Functions C) Discrete Functions D) Exponential Functions	1	1	2	2
9.	Identify the model that is best suited for sequential data A) Convolutional Neural Networks (ConvNets) B) Capsule Neural Networks (CapsNets) C) RNN (Recurrent Neural Network) D) Autoencoders	1	1	3	2

10. Convolutional Neural Network (CNN) is widely used for 1 1 3 2

A) Object detection
B) Image classification
C) Text generation
D) Reinforcement learning

11. State the purpose of pooling layer in a CNN 1 1 3 2

A) To reduce the spatial dimensions of the feature maps
B) To introduce non-linearity to the network
C) To adjust the weights and biases of the network
D) To compute the gradients for backpropagation

12. If your input image is 64x64x16, how many parameters are there in a single 1x1 convolution filter, including bias? 1 2 3 2

A) 2
B) 17
C) 4097
D) 1

13. Consider a Generative Adversarial Network (GAN) which successfully produces images of apples. Which of the following propositions is false? 1 2 4 13

A) The generator aims to learn the distribution of apple images.
B) The discriminator can be used to classify images as apple vs. non-apple.
C) After training the GAN, the discriminator loss eventually reaches a constant value
D) The generator can produce unseen images of apples

14. State the deep learning algorithm that is used for creating new data based on existing data. 1 1 4 13

A) Autoencoder
B) Deep Belief Network (DBN)
C) Generative Adversarial Network (GAN)
D) Variational Autoencoder (VAE)

15. Latent space is ------------- 1 2 4 13

A) compressed representation of the original data
B) extended representation of the original data
C) noise removed representation of the original data
D) both noise removed and extended representation of the original data

16. Which component of the VAE loss function ensures the latent variables adhere to a standard distribution? 1 2 4 2

A) Hinge loss
B) Absolute error
C) Cross-entropy
D) KL divergence

17. ImageNet uses ______ numbers of classes of objects 1 1 5 2

A) 1000
B) 10000
C) 100
D) more than 1 lakh

18. Resnet uses ------- 1 1 5 2

A) skipping connections
B) learning unreferenced functions
C) vanishing gradient decent
D) vanishing gradient accent

19. State the meaning of computer vision 1 1 5 13

A) The study of computer graphics
B) The study of computer networks
C) The study of algorithms for analyzing and interpreting visual data
D) The study of computer hardware architecture

20. Select the ImageNet benchmark from the given list 1 1 5 5

A) AlexNet
B) GAN
C) RNN
D) ReNet

PART - B (5 x 8 = 40 Marks)
Answer **ALL** Questions

	Question	Marks	BL	CO	PO
21 a.	Calculate the eigen value and eigen vector for the following matrix and give its importance. *1 1 3* *1 5 1* 3 1 1	8	3	1	2
	(OR)				
b.	Elaborate the working of Perceptron Learning algorithm with an example.	8	3	1	2
22 a.	Imagine that you are having more than 1,00,000 image datasets for training. Describe how early stopping works and why it's important to use it in conjunction with dropout.	8	3	2	2
	(OR)				
b.	Explain backpropagation algorithm with an example neural network.	8	3	2	2
23 a.	Summarize the difficulty of training in RNN algorithms and provide the possible solutions.	8	2	3	2
	(OR)				
b.	Describe the use of GRU and Compare with LSTM.	8	2	3	2
24 a.	Draw the basic architecture of Encoding and explain its working. Compare the encoders with PCA.	8	2	4	2
	(OR)				
b.	Discuss about the optimization algorithms in brief i)RMSprop for RNNs ii) SGD for CNN	8	2	4	5
25 a.	Explain the core idea of ResNet with its diagram.	8	2	5	2
	(OR)				
b.	Describe the concepts of image captioning with an example.	8	2	5	2

PART - C (1 x 15 = 15 Marks)
Answer **ANY ONE** Question

	Question	Marks	BL	CO	PO
26.	a) Illustrate the regularization methods. b) Apply perceptron learning model for solving XOR problem and explain it.	15	3	2	2
27.	a)Explain the concepts of CNN with a neat diagram. b) Explain Generative adversial network with a neat diagram.	15	2	4	5

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